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A MACHINE-LANGUAGE COMPUTER PROGRAM TO OBTAIN A NEUTRON SPECTRU--ETC(U)
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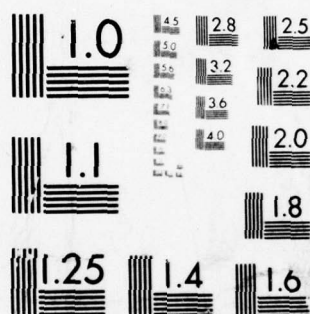
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A Machine-Language Computer Program to Obtain
a Neutron Spectrum from a Proton-Recoil Spectrum

September 1977

Computer Program to Obtain a Neutron Spectrum
from a Proton-Recoil Spectrum, by Craig R. Helmreich



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U.S. Army Material Development
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HARRY DIAMOND LABORATORIES
Adelphi, Maryland 20783

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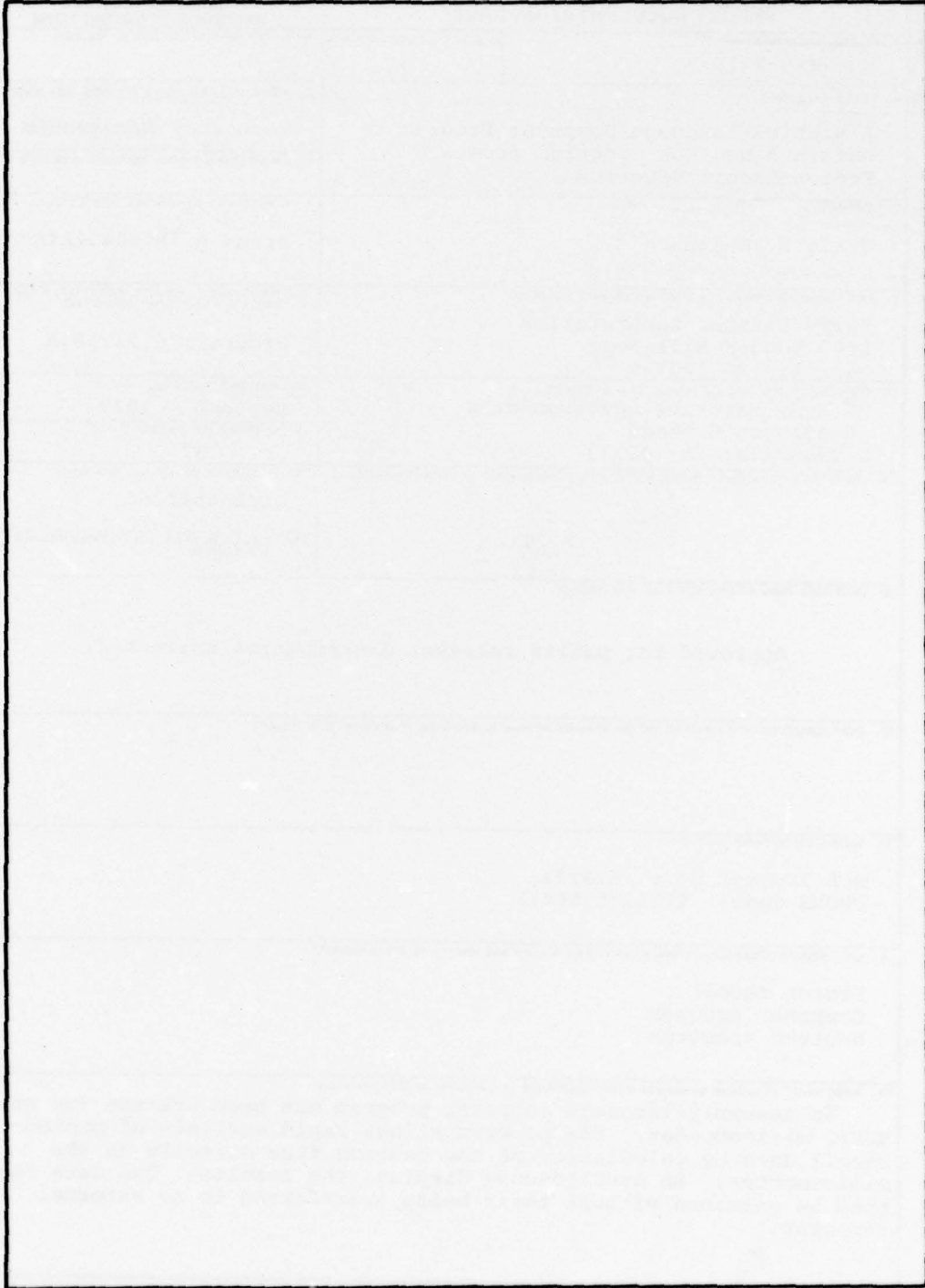
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1. INTRODUCTION

Fast-neutron spectrum measurements are important to studies of radiation vulnerability of Army electronic systems. The Harry Diamond Laboratories (HDL) is presently investigating the Bennett¹ neutron spectrometer, which determines the neutron spectrum by analysis of recoil protons in a hydrogen (H₂) or methane (CH₄) gas. The spectrometer is controlled by a minicomputer.

In data analysis, corrections must be made for carbon recoils and wall-and-end effects. The magnitudes of these corrections are not usually large, but are time-consuming to make. It was, therefore, decided to develop a quick method of obtaining preliminary results without these corrections. In this manner, the trend of the data could be followed as the data were taken, and preliminary spectrum results could be evaluated before the time-consuming final analysis of the data was undertaken.

2. STATEMENT OF PROBLEM

When neutrons below 14 MeV scatter from a detector with a gas containing protons, the energy spectrum of recoil protons is related to the energy spectrum of incident neutrons by

$$\phi_p(E) = N * T \int_0^E \frac{\sigma(E')}{E'} \phi_N(E') dE, \quad (1)$$

where N is the number density of hydrogen atoms in the detector gas, T is the collection time of the experiment, E is energy, and $\sigma(E)$ is the neutron-proton elastic scattering cross section. The proton flux is $\phi_p(E)$ and $\phi_N(E)$ is the neutron flux. It is necessary to differentiate equation (1) to obtain the desired neutron energy distribution.

As stated in the introduction, the various sources of background noise which are corrections to equation (1) are ignored at this stage of analysis. Of course, these corrections must eventually be made, but they are sufficiently time-consuming that the final results are not available for several weeks after the data have been taken.

3. METHOD OF SOLUTION

The proton-recoil spectrometer is controlled by an ND812 minicomputer. Since the minicomputer is always at the location where the data are taken, it is advantageous to use it for preliminary analysis. An assembly-language computer program to solve equation (1) for $\phi_N(E)$ was written for the ND812. It has the following features:

(a) Since the data are acquired and analyzed by the same computer, there is no need to transfer data from one machine to another. Since the only output device from the ND812 (compatible with other computers) is paper tape, any transfer of data would be time-consuming.

(b) Since the ND812 is owned by HDL, there is no data-processing expense.

¹ E. F. Bennett and T. J. Yule, Techniques and Analyses of Fast-Reactor Neutron Spectroscopy with Proton-Recoil Proportional Counters, Argonne National Laboratory, ANL-7763 (1971).

(c) The system provides a visual display of both neutron and proton spectra on an oscilloscope. A listing of the data is also available.

(d) The analysis is done in a matter of minutes, so that any problems associated with the data may be seen as the data are taken, rather than weeks later. If there are problems, adjustments may be made immediately, without wasting days of valuable reactor running time.

4. PROGRAM DESCRIPTION

The equation relating the neutron spectrum $\phi_N(E)$ to the derivative of the proton-recoil distribution $d\phi_p(E)/dE$ is

$$\phi_N(E) = - \frac{1}{N \cdot T} \frac{E}{\sigma(E)} \frac{d\phi_p(E)}{dE} \quad (2)$$

The data $\phi_p(E)$ consist of the number of recoil protons detected for each of several energy intervals of equal size. Numerical differentiation is a noise-generating process, and the data contain statistical fluctuations, so a smoothing process is used to obtain the best results. The method used is that of Bennett,² who suggests that the slope of a least-squares line—fit through the data surrounding a data point—be used as the slope of the data. The width of the interval over which the least-squares line is fitted is

$$STHW = FUDG \cdot \sqrt{WI^2 + CON/E} \quad (3)$$

where STHW is the half-width of the slope-taking interval. WI is the intrinsic resolution of the detector, and the constant, CON, adds a factor to account for the statistical nature of the gas amplification. FUDG is an arbitrary scale factor which allows more smoothing. Bennett² demonstrates that a FUDG factor as large as 3.0 does not significantly affect the area under a peak, although significant line broadening does occur at that value. A value of 2.0 to 3.0 for FUDG is found to give the best compromise between smoothing and loss of resolution.

The program (listed in app A) is written in BASC-12, the assembly language supplied by the computer manufacturer.³ The program consists of a main control routine, which calls various subroutines as they are needed, and the subroutines. A package⁴ supplied by the manufacturer is used to do the floating-point arithmetic since hard-wired floating-point arithmetic is not available.

In general, the program functions in the following manner. A subroutine INPT is called to obtain the calibration data and constants needed to run the program. The program then locates the proton-recoil data in core and a least-squares line is fitted through the intervals determined by equation (3). For each energy interval, the slope is used in equation (2) to calculate the neutron flux spectrum. The results are stored in the minicomputer memory and may be viewed on the oscilloscope or output to Teletype or tape cassette by use of the ND-1075-01 computer-program.⁵

² E. F. Bennett, Fast Neutron Spectroscopy by Proton-Recoil Proportional Counters, Nuclear Science Engineering 27(1967), 16.

³ Nuclear Data, Incorporated, Software Instruction Manual, BASC-12 General Assembler, Palatine, IL (1971).

⁴ Nuclear Data, Incorporated, ND812 Utilities Manual, Palatine, IL (1971), Ch 9 through 12.

⁵ Nuclear Data, Incorporated, Software Instruction Manual, ND4420 Single/Dual Parameter Monitor, Palatine, IL (1972).

4.1 Description of Main Subroutines

INPT is the routine which reads the calibration data, along with N, T, WI, and FUDG (See eq (1) and (2).) Also, the IN and OUT groups must be input to INPT. These input numbers tell the computer where in memory to locate the proton-recoil data and where to put the neutron flux results.

INIT takes the calibration data and defines the energy scale.

IOST accepts the input and output group numbers and translates them into locations in core.

BUFRT computes the slope-taking half width and transfers the appropriate proton-recoil data to a buffer location. It is possible for equation (3) to give a width which reaches beyond the data. If this occurs, the slope-taking interval is reduced so that it reaches only to the end of the data.

LSR is the least-squares fitting routine. The sums necessary for the fit are accumulated in integer (as opposed to floating-point) format to save running time. A special routine, TPCHK, converts the sums necessary for least-squares fitting from integer to floating-point format, since they may exceed the normal double-precision 24-bit length of an integer in the computer. The result of program LSR is the slope of the least-squares line.

OPT computes the flux $\phi_N(E)$ from the slope using equation (2). The result is stored as an integer in the memory of the computer, where it may later be accessed by the ND-1075-01.

4.2 Use of Program

First, the ND-1075-01 program must be loaded into core. This program is used to load the raw proton-recoil data. The data may be loaded into groups of any size, as the derivative program automatically adjusts itself to fit the group size specified by the ND-1075-01. However, the data may not be loaded into the buffer memory extension, since the derivative program has access only to fields 02 and 03 of core.

Next, the floating-point package is loaded into field 01, along with the SQRT and operate instruction overlays.

The derivative program is loaded in field 01. The starting address is 4000 F01. Field 00 is not altered, so that the functions supported by the 1075-01 located in field 00 are not affected. These functions include all display- and data-acquisition functions but not I/O routines or any functions which use the Teletype.

A sample run of the program is included in appendix B. First, two points and their corresponding energies are input. They set the energy scale of the data. Then, the count time of the experiment, the hydrogen atom concentration of the proton-recoil tube, and mechanical width of the tube are input for use in equations (1) and (2). Next are given the input group and output group, which tell the program where to get and to put the processed data. The input group may not equal the output group, and neither may access the buffer memory extension. Finally, the slope-taking multiplier is input for use in equation (3).

The input data may take various floating-point forms. For example, the number "one" may be input as 1.0, 1, or 1.0E+00. If, after a number is typed, the space bar is pushed to terminate that number, the program will continue typing on the same line. If a carriage return is pushed, the typing will proceed from the beginning of the next line.

When the derivative program is finished, it returns control automatically to the 1075-01. The results may then be viewed on the oscilloscope. If the derivative of another group of data is to be determined, the 1075-01 must be stopped and the address register set again to 4000 F01. In this manner, the derivative program may be run any number of times in succession. However, if the data are to be output, it is necessary to reload and use the 1075-01 program. A typical running time for the program is 15 to 20 s for 256 points.

5. SUMMARY

A rapid, assembly-language computer program has been written to derive neutron spectra from proton-recoil spectra. The program allows a preliminary analysis of data to be made without having to rely on an external computer. The program is an invaluable aid to on-line evaluation of the quality of data during acquisition.

LITERATURE CITED

1. E. F. Bennett and T. J. Yule, *Techniques and Analyses of Fast-Reactor Neutron Spectroscopy with Proton-Recoil Proportional Counters*, Argonne National Laboratory, ANL-7763 (1971).
2. E. F. Bennett, *Fast Neutron Spectroscopy by Proton-Recoil Proportional Counters*, *Nuclear Science Engineering* 27(1967), 16.
3. Nuclear Data, Incorporated, *Software Instruction Manual, BASC-12 General Assembler*, Palatine, IL (1971).
4. Nuclear Data, Incorporated, *ND812 Utilities Manual*, Palatine, IL (1971), Ch 9 through 12.
5. Nuclear Data, Incorporated, *Software Instruction Manual, ND4420 Single/Dual Parameter Monitor*, Palatine, IL (1972).

APPENDIX A.—The BASC-12 Computer Program

This program was written for the ND812 minicomputer language, BASC-12, supplied by the manufacturer.¹ The program consists of a main control routine. The subroutines are called as needed. Since hard-wired floating-point arithmetic is not available, a package is supplied by the manufacturer for the floating-point arithmetic.

```
FSIP=1007
FCLR=1004
FSORT=7401
FSIN=1006
FJMP=6000
FNEG=1003
IFIX=7405
EXJK=1374
FLCAT=7406
XMAX=1131          /REFERS TO ND-1075 GROUP SIZE
```

[FIELD 01

*2540

/MODIFY FLOATING POINT PACKAGE OUTPUT ROUTINE SO IT WORKS

```
/
2540  7413      TCP      /OUTPUT CHARACTER
2541  7414      TOS
2542  6101      JMP  .-1 /WAIT TILL READY
```

*4000

/CONTROL PORTION OF PROGRAM
/THIS PORTION CALLS OTHER ROUTINES AS NEEDED

```
/
4000  0640      TWJPS
4001  4071      INPT     /INPUT DATA
4002  0640      TWJPS
4003  4770      INIT     /PREPARE EFP
4004  0640      TWJPS
4005  4371      IOST     /LOCATE IN AND OUT GROUPS
4006  0504      TWLDJ FO
4007  1131      XMAX     /SET COUNTER
```

¹ Nuclear Data, Incorporated, Software Instruction Manual, BASC-12 General Assembler, Palatine, IL (1971).

APPENDIX A

```

4010 2301          SUBL 01 /SKIP FIRST AND LAST POINTS
4011 5422          STJ CNT1
4012 1510          CLR J
4013 2202          ADDL 02 /FIRST POINT WHICH IS DONE
4014 5420          STJ CNT2
4015 5017 CON,     LDJ CNT2          /TAKE DERIV ONE POINT AT A TIME
4016 0540          TWSTJ
4017 4567          PTNO
4020 0640          TWJPS
4021 4575          BUFRT          /PUT DATA INTO BUFFER
4022 0640          TWJPS
4023 5146          LSR          /LEAST SQUARE FIT
4024 0640          TWJPS
4025 4473          OPT          /PUT FLUX IN OUTPUT GROUP
4026 3406          ISZ CNT2
4027 3004          DSZ CNT1          /LAST POINT?
4030 6113          JMP CON /NO
4031 0604          TWJMP FO
4032 0200          0200          /YES, RETURN TO 1075

4033 0000 CNT1,    0000          /COUNTER
4034 0000 CNT2,    0000          /POINT NO

/OUTPUT MESSAGE ROUTINE
4035 0000 OMR,     0000
4036 5301          LDJ0 OMR          /GET START ADDRESS OF MESSAGE
4037 3502          ISZ OMR
4040 5411          STJ MESSG
4041 5210 OMR1,    LDJ0 MESSG          /GET CHARACTER
4042 1501          SNZ J          /END OF MESSAGE?
4043 6306          IRETURN OMR          /YES
4044 7413          TCP          /NO, TYPE IT
4045 7414          TOS
4046 6101          JMP .-1
4047 3402          ISZ MESSG
4050 6107          JMP OMR1          /GET NEXT CHAR
4051 0000 MESSG,   0000

4052 0000 ADDR,    0000          /WHERE TO PUT DATA
4053 1400 ADRI,    1400
4054 0000 STT,     0000          /INPUT THRU FLOAT POINT PACK
4055 5102          LDJ ADRI          /PREP TO ENTER FPP
4056 0540          TWSTJ
4057 0402          0402
4060 0640          TWJPS
4061 0400          0400          /ENTER FPP
4062 1400          FINPUT
4063 5711          FSTORE ADDR
4064 1001          FEXIT
4065 5113          LDJ ADDR          /NEXT DATA GOES THREE WORDS UP
4066 2203          ADDL 03

```

```

4067 5515      STJ ADDR
4070 6314      [RETURN STT

/WRITE TITLES
/GET DATA THRU FLOATING POINT PACKAGE
/
4071 0000  INPT,  0000
4072 0640  A1,    TWJPS
4073 4035      OMR      /OUTPUT MESSAGE
4074 4232      TITLE   /DERIVATIVE PROGRAM
4075 7103      XCT A1
4076 4211      POINT    /POINT=
4077 5053      LDJ DAT1      /POINT TO START OF DATA BUFFER
4100 5526      STJ ADDR
4101 6525      JPS STT /GET POINT
4102 7110      XCT A1 /PRINT MESSAGE
4103 4221      ENE      /ENERGY =
4104 6530      JPS STT /GET ENERGY
4105 7113      XCT A1
4106 4211      POINT    /POINT=
4107 6533      JPS STT
4110 7116      XCT A1
4111 4221      ENE      /ENERGY=
4112 6536      JPS STT
4113 7121      XCT A1
4114 4257      CTM      /COUNT TIME
4115 6541      JPS STT
4116 7124      XCT A1
4117 4274      H2CO      /H CONCENTRATION
4120 6544      JPS STT
4121 7127      XCT A1
4122 4306      MEW      /ME WIDTH
4123 6547      JPS STT
4124 7132      XCT A1
4125 4322      DFRM      /DATA FROM GRP
4126 6552      JPS STT
4127 7135      XCT A1
4130 4335      DTO      /DATA TO GROUP
4131 6555      JPS STT
4132 7140      XCT A1
4133 4354      FUDG      /MULTIPLIER
4134 6560      JPS STT
4135 5162      LDJ ADRI      /PREP TO ENTER FPP
4136 0540      TWSTJ
4137 0402      0402
4140 0640      TWJPS
4141 0400      0400
4142 5036      FLOAD GF      /CHANGE GF,GT TO INTEGERS
4143 7405      IFIX
4144 5434      FSTOR GF
4145 5036      FLOAD GT
4146 7405      IFIX
4147 5434      FSTOR GT

```

APPENDIX A

```
4150 1001 FEXIT
4151 6360 [RETURN INPT
```

```
/END OF BLOCK 1
//
```

```
IC AT 4153
IC AT 4153
##
```

```
4152 4153 DAT1, P1
4153 0000 P1, 0000 /INPUT DATA BUFFER
4154 0000 0000
4155 0000 0000
4156 0000 E1, 0000
4157 0000 0000
4160 0000 0000
4161 0000 P2, 0000
4162 0000 0000
4163 0000 0000
4164 0000 E2, 0000
4165 0000 0000
4166 0000 0000
4167 0000 CT, 0000 /COUNT TIME
4170 0000 0000
4171 0000 0000
4172 0000 HC, 0000 /H CONCENTRATION
4173 0000 0000
4174 0000 0000
4175 0000 MW, 0000 /MECH WIDTH
4176 0000 0000
4177 0000 0000
4200 0000 GF, 0000 /INPUT GROUP
4201 0000 0000
4202 0000 0000
4203 0000 GT, 0000 /OUTPUT GROUP
4204 0000 0000
4205 0000 0000
4206 0000 FDGE, 0000 /WI MULTIPLIER
4207 0000 0000
4210 0000 0000

4211 0320 POINT, 0320 /POINT
4212 0317 0317
4213 0311 0311
4214 0316 0316
4215 0324 0324
4216 0240 0240
4217 0275 0275
4220 0000 0000

4221 0305 ENE, 0305 /ENERGY
4222 0316 0316
4223 0305 0305
```

4224	0322	0322
4225	0307	0307
4226	0331	0331
4227	0240	0240
4230	0275	0275
4231	0000	0000

4232	0304	TITLE,	0304	/TITLE
4233	0305		0305	
4234	0322		0322	
4235	0311		0311	
4236	0326		0326	
4237	0301		0301	
4240	0324		0324	
4241	0311		0311	
4242	0326		0326	
4243	0305		0305	
4244	0240		0240	
4245	0320		0320	
4246	0322		0322	
4247	0317		0317	
4250	0307		0307	
4251	0322		0322	
4252	0301		0301	
4253	0315		0315	
4254	0215		0215	
4255	0212		0212	
4256	0000		0000	

4257	0303	CTM,	0303	/COUNT TIME
4260	0317		0317	
4261	0325		0325	
4262	0316		0316	
4263	0324		0324	
4264	0240		0240	
4265	0324		0324	
4266	0311		0311	
4267	0315		0315	
4270	0305		0305	
4271	0240		0240	
4272	0275		0275	
4273	0000		0000	

4274	0310	H2CO,	0310	/H CONCEN
4275	0240		0240	
4276	0303		0303	
4277	0317		0317	
4300	0316		0316	
4301	0303		0303	
4302	0305		0305	
4303	0316		0316	
4304	0275		0275	
4305	0000		0000	

APPENDIX A

4306	0315	MEW,	0315	/ME WIDTH
4307	0305		0305	
4310	0303		0303	
4311	0310		0310	
4312	0240		0240	
4313	0327		0327	
4314	0311		0311	
4315	0304		0304	
4316	0324		0324	
4317	0310		0310	
4320	0275		0275	
4321	0000		0000	

4322	0311	DFRM,	0311	/INPUT GROUP
4323	0316		0316	
4324	0240		0240	
4325	0307		0307	
4326	0322		0322	
4327	0317		0317	
4330	0325		0325	
4331	0320		0320	
4332	0240		0240	
4333	0275		0275	
4334	0000		0000	

4335	0317	DTO,	0317	/OUT GROUP
4336	0325		0325	
4337	0324		0324	
4340	0000		00	
4341	0325		0325	
4342	0324		0324	
4343	0240		0240	
4344	0307		0307	
4345	0322		0322	
4346	0317		0317	
4347	0325		0325	
4350	0320		0320	
4351	0240		0240	
4352	0275		0275	
4353	0000		0000	

4354	0327	FUDG,	0327	/WI MULT
4355	0311		0311	
4356	0240		0240	
4357	0315		0315	
4360	0325		0325	
4361	0314		0314	
4362	0324		0324	
4363	0275		0275	
4364	0240		0240	
4365	0000		0000	

/END OF BLOCK 2
//

```

IS C%      AT 4366
4366 0000 ;%
/TRANSLATE IN GROUP AND OUT GROUP NUMBERS
/ INTO ACTUAL LOCATION IN CORE
/
4367 0604 RETN,   TWJMP FO
4370 0200                0200

4371 0000 IOST,   0000      /LOCATE GROUPS IN CORE
4372 0500                TWLDJ
4373 4202                GF+2
4374 0510                TWLDK   /IN GROUP = OUT GROUP?
4375 4205                GT+2
4376 1131                NSJK J
4377 1501                SNZ J
4400 6111                JMP RETN      /YES, RETURN
4401 0514                TWLDK FO      /NO, SET INPUT DATA
4402 1131                XMAX
4403 1604                INC K   /NO OF PTS PER GROUP
4404 0500                TWLDJ
4405 4202                GF+2
4406 2301                SUPL 01 /NO OF GROUPS - 1
4407 1161                ROTD J 01   /DOUBLE PRECISION
4410 1000                MPY
4411 1302                LJKFRS
4412 0540                TWSTJ
4413 4572                INST      /LOCATION OF INPUT GROUP
4414 1604                INC K
4415 1604                INC K   /DATA STARTS IN FIELD 2
4416 0550                TWSTK
4417 4573                INST+1 /FIELD
4420 0514                TWLDK FO      /DO SAME FOR OUTPUT GROUP
4421 1131                XMAX
4422 1604                INC K
4423 0500                TWLDJ
4424 4205                GT+2
4425 2301                SUPL 01
4426 1161                ROTD J 01
4427 1000                MPY
4430 1302                LJKFRS
4431 0540                TWSTJ
4432 4462                OUST      /LOCATION OF OUTPUT GROUP
4433 1604                INC K
4434 1604                INC K
4435 0550                TWSTK
4436 4463                OUST+1
4437 6346                IRETURN IOST

```

/CONSTANTS FOR HYDROGEN CROSS SECTION
4440 0016 CON1, 0016 /11010.

APPENDIX A

4441	2540		2540	
4442	2000		2000	
4443	0014	CON2,	0014	/4041.
4444	3744		3744	
4445	4000		4000	
4446	0014	CON3,	0014	/2387.
4447	2251		2251	
4450	4000		4000	
4451	0010	CON4,	0010	/135.5
4452	2074		2074	
4453	0000		0000	
4454	0000	STR1,	0000	/GENERAL STORAGE
4455	0000		0000	
4456	0000		0000	
4457	0000	STR2,	0000	/GENEL STORAGE
4460	0000		0000	
4461	0000		0000	
4462	0000	OUST,	0000	/WHERE TO PUT OUTPUT DATA
4463	0000		0000	/FIELD
4464	0000	FLUX,	0000	
4465	0000		0000	
4466	0000		0000	
4467	1400	CON5,	1400	
4470	0000	SIG,	0000	/CROSS SECTION
4471	0000		0000	
4472	0000		0000	

/CONVERT SLOPE TO FLUX AND
/ PUT INTO CORRECT OUTPUT LOCATION
/

4473	0000	OPT,	0000	
4474	5105		LDJ CON5	/COMPUTE FLUX
4475	0540		TWSTJ	
4476	0402		0402	
4477	0640		TWJPS	
4500	0400		0400	/ENTER FLOAT POINT PACK
4501	0500		FTWLD	/COMPUTE H CROSS SECTION
4502	5034		E	
4503	4540		FADD CON2	
4504	5530		FSTOR STR1	
4505	0500		FTWLD	
4506	5034		E	
4507	4536		FADD CON4	
4510	5531		FSTOR STR2	
4511	5151		FLOAD CON1	
4512	6536		FDIV STR1	
4513	5537		FSTOR STR1	
4514	5146		FLOAD CON3	
4515	6536		FDIV STR2	
4516	4542		FADD STR1	
4517	5527		FSTOR SIG	/STORE IT

APPENDIX A

4577	0540	TWSTJ
4600	5033	P+2
4601	0640	TWJPS
4602	5045	EFP /FIND ENERGY
4603	0640	TWJPS
4604	4734	WIN /FIND WIDTH
4605	0500	TWLDJ
4606	4724	WIP+2 /SLOPE TAKING HALF INTERVAL
4607	5516	STJ PTWTH
4610	4121	SBJ PTNO
4611	2301	SUBL 01
4612	1502	SIP J /INTERVAL TOO WIDE (LOW)?
4613	6004	JMP .+4 /NO
4614	5125	LDJ PTNO /YES,
4615	2301	SUBL 01
4616	5525	STJ PTWTH /DO FROM FIRST POINT
4617	0504	PT2, TWLDJ FO
4620	1131	XMAX
4621	4132	SBJ PTNO
4622	4131	SBJ PTWTH
4623	1506	SIN J /TOO WIDE(HIGH)?
4624	6004	JMP .+4 /NO
4625	7106	XCT PT2 /YES, DO TO MAX
4626	4137	SBJ PTNO
4627	5536	STJ PTWTH
4630	5141	LDJ PTNO /FIND PFST
4631	4140	SBJ PTWTH
4632	5542	STJ PFST
4633	5142	LDJ PTWTH /FIND NO OF POINTS IN INTERVAL
4634	1161	ROTD J 01
4635	1504	INC J
4636	5542	STJ CNTR
4637	0540	TWSTJ
4640	5230	NPTS /FOR LSR
4641	5013	LDJ PT4 /SET FIELD BITS
4642	2004	ANDF MASK
4643	4550	ADJ INST+1
4644	5410	STJ PT4
4645	6002	SKIP
4646	7774	MASK, 7774
4647	0640	TWJPS
4650	4673	PFSET /TRANSLATE PFST TO LOCATION IN CORE
4651	0500	TWLDJ
4652	4566	BUFF
4653	5400	STJ FIRST
4654	0524	PT4, TWLDJ FO /LOAD DATA - DOUBLE PRECISION
4655	4570	PFST
4656	5600	STJ FIRST /PUT INTO BUFFER
4657	3567	ISZ PFST
4660	6002	SKIP
4661	3505	PT5, ISZ PT4
4662	7106	XCT PT4
4663	5600	STJ FIRST

APPENDIX A

4520	0500		FTWLD	/COMPUTE FLUX
4521	5034		E	/FLUX=-(E/(N*SIG*CT)) * (SLOPE/DEPP)
4522	6532		FDIV SIG	
4523	0700		FTWMT	
4524	5416		SLOPE	
4525	0640		FTWDV	
4526	4172		HC	
4527	0640		FTWDV	
4530	4167		CT	
4531	0640		FTWDV	
4532	5037		DEPP	
4533	1003		FNEG	
4534	1007		FSIP	/FLUX < 0 ?
4535	1004		FCLR	/YES, FLUX = 0
4536	7405		IFIX	
4537	5553		FSTOR FLUX	
4540	1001		FEXIT	
4541	5017		LDJ PT3	/GET FIELD
4542	4557		ADJ OUST+1	
4543	5402		STJ PT1	
4544	5156		LDJ FLUX+2	/GET FLUX (LOW ORDER)
4545	0564	PT1,	TWSTJ0 FO	/STORE IT
4546	4462		OUST	
4547	3565		ISZ OUST	
4550	6002		SKIP	
4551	6410		JPS INCR	
4552	5165		LDJ FLUX+1	/GET FLUX (HIGH ORDER)
4553	7106		XCT PT1	/STORE IT
4554	3572		ISZ OUST	
4555	6002		SKIP	
4556	6403		JPS INCR	
4557	6364		[RETURN OPT	
4560	0564	PT3,	TWSTJ0 FO	
4561	0000	INCR,	0000	/INCREMENT FIELD
4562	0340		TWISZ	
4563	4463		OUST+1	
4564	3517		ISZ PT1	
4565	6304		[RETURN INCR	
4566	7000	BUFF,	7000	/BUFFER LOCATION
4567	0000	PTNO,	0000	/NO OF POINT IN QUESTION
4570	0000	PFST,	0000	/FIRST POINT OF SLOPE-TAKING INTERVAL
4571	0000	PTWTH,	0000	/HALF WIDTH
4572	0000	INST,	0000	/DATA LOCATION IN MEMORY
4573	0000		0000	/FIELD
4574	0000	CNTR,	0000	/NO OF POINTS TO TRANSFER
/TRANSFER POINTS IN SLOPE-TAKING INTERVAL INTO BUFFER				
/				
4575	0000	BUFRT,	0000	
4576	5107		LDJ PTNO	

APPENDIX A

4664	3574		ISZ PFST	
4665	6002		SKIP	
4666	3512		ISZ PT4	
4667	3173		DSZ CNTR	/LAST PT?
4670	6114		JMP PT4	/NO
4671	6374		[RETURN BUFRT	/YES

4672	4661	ADR5,	PT5	
4673	0000	PFSET,	0000	/CONVERT POINT NO TO CORE LOCATION
4674	1710		CLR JK	
4675	0500		TWLDJ	
4676	4570		PFST	/GET NUM OF FIRST POINT
4677	2301		SUBL 01	
4700	1361		ROTD JK 01	/DOUBLE PRECISION
4701	1450		CLR 0	
4702	0440		TWADJ	
4703	4572		INST	/CONVERT TO CORE LOCATION
4704	0540		TWSTJ	
4705	4570		PFST	/FIRST CORE LOCATION TO GET
4706	1445		SIZ 0	/CHECK FIELD
4707	7315		XCT# ADR5	
4710	1605		SIZ K	
4711	7317		XCT# ADR5	
4712	6317		[RETURN PFSET	

/END OF BLOCK 3

//

IC AT 4714
 IS C AT 4713
 4713 0000 #;

4714	7776	CONST1,	7776	/0.17
4715	2560		2560	
4716	5076		5076	
4717	0000	WI,	0000	/INTRINSIC WIDTH (KEV)
4720	0000		0000	
4721	0000		0000	
4722	0000	WIF,	0000	/WI IN POINTS (HALF WIDTH)
4723	0000		0000	
4724	0000		0000	
4725	0000	WMSQ,	0000	/WM*WM
4726	0000		0000	
4727	0000		0000	
4730	1400	FPCON1,	1400	
4731	0002	TWO,	0002	/FLOATING POINT 2.0
4732	2000		2000	
4733	0000		0000	

/FIND SLOPE TAKING HALF INTERVAL

/

4734	0000	WIN,	0000	/WI=SQRT(WM**2+CONST1/E)
4735	5105		LDJ FPCON1	

APPENDIX A

4736	0540		TWSTJ	
4737	0402		0402	
4740	0640		TWJPS	
4741	0400		0400	/ENTER FLOAT POINT PACK
4742	5126		FLOAD	CONST1
4743	0640		FTWDV	
4744	5034		E	
4745	0440		FTWAD	
4746	4725		WMSQ	
4747	7401		FSORT	
4750	0700		FTWMT	
4751	5034		E	
4752	5533		FSTOR	WI /WI IN KEV
4753	0640		FTWDV	
4754	5037		DEPP	/WI IN POINTS
4755	0700		FTWMT	
4756	4206		FDGE	
4757	6526		FDIV	TWO /HALF INTERVAL
4760	7405		IFIX	
4761	5537		FSTOR	WIP /WIDTH IN POINTS
4762	1001		FEXIT	
4763	6327		IRETURN	WIN
4764	0000	STOR1,	0000	
4765	0000		0000	/FLOAT NO TEMP STORAGE
4766	0000		0000	
4767	1400	ADR6,	1400	
4770	0000	INIT,	0000	/INITIALIZE S0
4771	5102		LDJ	ADR6 /FLOATING CONSTANTS
4772	0540	EN1,	TWSTJ	
4773	0402		0402	
4774	0640	EN2,	TWJPS	
4775	0400		0400	
4776	0500		FTWLD	/FIND DEPP FROM CALIBRATION POINTS
4777	4161		P2	
5000	0400		FTWSB	
5001	4153		P1	
5002	5516		FSTOR	STOR1
5003	0500		FTWLD	
5004	4164		E2	
5005	0400		FTWSB	
5006	4156		E1	
5007	6523		FDIV	STOR1
5010	5427		FSTOR	DEPP
5011	0500		FTWLD	/FIND B FOR E=PT*DEPP+B
5012	4153		P1	
5013	7024		FMULT	DEPP
5014	5530		FSTOR	STOR1
5015	0500		FTWLD	
5016	4156		E1	

APPENDIX A

```

5017 4133      FSUB STOR1
5020 5422      FSTOR B
5021 0500      FTWLD /INITIALIZE WMSQ
5022 4175      MW
5023 0700      FTWMT
5024 4175      MW
5025 0540      FTWST
5026 4725      WMSQ
5027 1001      FEXIT
5030 6340      [RETURN INIT

5031 0000 P,    0000 /POINTER STORAGE
5032 0000      0000
5033 0000      0000
5034 0000 E,    0000 /ENERGY RESULT
5035 0000      0000
5036 0000      0000
5037 0000 DEPP, 0000 /E=DEPP X P + B
5040 0000      0000
5041 0000      0000
5042 0000 B,    0000
5043 0000      0000
5044 0000      0000

5045 0000 EFP,  0000 /FIND ENERGY FROM POINT
5046 5157      LDJ ADR6
5047 7155      XCT EN1
5050 7154      XCT EN2 /ENTER FPP
5051 5120      FLOAD P
5052 7406      FLOAT
5053 7114      FMULT DEPP
5054 4512      FADD B
5055 5521      FSTOR E /E=DEPP*P+B
5056 1001      FEXIT
5057 6312      [RETURN EFP

/XCLR STORES ZEROS IN SUCCESSIVE LOCATIONS
/CALL: JPS XCLR
/      START- FIRST WORD CLEARED
/      NO - NO OF WORDS CLEARED
/      RETURN POINT
/
5060 0000 XCLR, 0000
5061 5301      LDJ0 XCLR
5062 3502      ISZ XCLR
5063 5406      STJ AD1 /FIRST WD TO BE ZEROED
5064 5304      LDJ0 XCLR
5065 5411      STJ XC2 /NO OF WDS TO BE ZEROED
5066 3506      ISZ XCLR /SET RETURN ADDRESS
5067 1510      CLR J
5070 0540      TWSTJ
5071 0000 AD1,  0000

```


APPENDIX A

```
5072 3501      ISZ AD1
5073 3003      DSZ XC2
5074 6104      JMP .-4
5075 6315      [RETURN XCLR
5076 0000 XC2, 0000
```

```
/END OF BLOCK 4
//
```

IC AT 5100

5077 0003 #3

/CONVERT DOUBLE OR TPLE PRECISION INTEGER

/TO FLOATING POINT

/

```
5100 0000 TPCHK, 0000
5101 0500      TWLDJ
5102 5407      POSST
5103 5437      STJ POS /LOCATION OF NUMBER
5104 2302      SUPL 02
5105 5434      STJ POS2
5106 0500      TWLDJ /PREPARE FPP
5107 5254      IDCON1
5110 0540      TWSTJ
5111 0402      0402
5112 0640      TWJPS
5113 0400      0400 /ENTER IT
5114 0520      FTWLD@ /GET LOW ORDER TWO WORDS OF NUMBER
5115 5142      POS
5116 7406      FLOAT
5117 5424      FSTOR STOR5
5120 1006      FSIN /IF NEG, NO WANTED=PREC+NO FOUND
5121 6004      FJMP .+4 /SINCE ND ASSUMES DOUBLE PREC INTEGE
5122 5014      FLOAD PREC /WITH HIGH ORDER BIT SET TO BE NEG
5123 4420      FADD STOR5
5124 5417      FSTOR STOR5
5125 0520      FTWLD@
5126 5141      POS2 /GET HIGH ORDER PART OF NUM
5127 7406      FLOAT
5130 7006      FMULT PREC /CORRECT MAGNITUDE
5131 4412      FADD STOR5 /ADD LOW ORDER PART
5132 0560      FTWST@ /STORE FLOATING NUMBER
5133 5142      POS
5134 1001      FEXIT
5135 6335      [RETURN TPCHK

5136 0031 PREC, 0031 /4096*4096
5137 2000      2000
5140 0000      0000
5141 0000 POS2, 0000
5142 0000 POS, 0000
5143 0000 STOR5, 0000
5144 0000      0000
5145 0000      0000
```

APPENDIX A

```

/LINEAR LEAST SQUARES FIT ROUTINE
/ACCUMULATE SUMS IN INTEGER FORMAT TO SAVE TIME
/
/
5146 0000 LSR, 0000
5147 0640 TWJPS
5150 5060 XCLR /CLEAR SUMS
5151 5231 ICON1
5152 0021 0021
5153 1504 INC J /INIT POINTERS
5154 5477 STJ ACNT2 /AND COUNTERS
5155 5053 LDJ NPTS
5156 5474 STJ ACNT1
5157 5472 STJ N+2
5160 0500 TWLDJ
5161 4566 BUFF
5162 5400 STJ FIRST
5163 5070 CONTN2, LDJ ACNT2 /SUM OF I
5164 1450 CLR 0
5165 4446 ADJ ICON1+2
5166 5445 STJ ICON1+2
5167 1445 SIZ 0 /OVERFLOW?
5170 3442 ISZ ICON1+1 /YES
5171 5062 LDJ ACNT2 /NO, SUM I X I
5172 1204 LKFJ
5173 1000 MPY /I X I
5174 1302 LJKFRS
5175 1450 CLR 0
5176 4440 ADJ ICON2+2 /LOW ORDER
5177 5437 STJ ICON2+2
5200 1445 SIZ 0 /OVERFLOW?
5201 1604 INC K /YES
5202 1374 EXJK /NO
5203 4432 ADJ ICON2+1
5204 5431 STJ ICON2+1
5205 5200 LDJ0 FIRST /Y LOW
5206 1374 EXJK
5207 5200 LDJ0 FIRST /Y HIGH
5210 1374 EXJK
5211 1301 LRSFJK /STORE DATA
5212 1450 CLR 0
5213 4433 ADJ Y+2 /SUM ON Y
5214 5432 STJ Y+2
5215 1445 SIZ 0 /CHECK FOR CARRIES TO HIGHER PRECISION
5216 3427 ISZ Y+1
5217 6002 SKIP
5220 3424 ISZ Y
5221 1450 CLR 0
5222 1374 EXJK
5223 4422 ADJ Y+1
5224 5421 STJ Y+1
5225 1445 SIZ 0
5226 3416 ISZ Y
5227 6030 JMP CONTN1

```

APPENDIX A

5230	0000	NPTS,	0000	/NO OF POINTS
5231	0000	ICON1,	0000	/SUM OF I
5232	0000		0000	
5233	0000		0000	
5234	0000	ICON2,	0000	/SUM OF I X I
5235	0000		0000	
5236	0000		0000	
5237	0000		0000	
5240	0000	IY,	0000	/SUM OF I X Y
5241	0000		0000	
5242	0000		0000	
5243	0000		0000	
5244	0000	Y,	0000	/SUM OF Y
5245	0000		0000	
5246	0000		0000	
5247	0000	N,	0000	/NO OF POINTS
5250	0000		0000	
5251	0000		0000	
5252	0000	ACNT1,	0000	/COUNTER
5253	0000	ACNT2,	0000	/I
5254	1400	IDCON1,	1400	
5255	0600	CJMP,	TWJMP	
5256	5163		CONTN2	
5257	1302	CONTN1,	LJKFRS	/RESTORE Y
5260	0550		TWSTK	/FIND I X Y
5261	5410		STOR6	/HIGH ORDER LAST
5262	0510		TWLDK	
5263	5253		ACNT2	/I
5264	1000		MPY	/I X LOW ORDER - DOUBLE PRECISION
5265	1302		LJKFRS	
5266	1450		CLR 0	
5267	4525		ADJ IY+2	/ADD IT (LOW)
5270	5526		STJ IY+2	/STORE IT
5271	1374		EXJK	
5272	1445		SIZ 0	/CHECK FOR CARRIES
5273	3532		ISZ IY+1	
5274	6002		SKIP	
5275	3535		ISZ IY	
5276	1450		CLR 0	
5277	4536		ADJ IY+1	/ADD IT (HIGH)
5300	5537		STJ IY+1	
5301	1445		SIZ 0	
5302	3542		ISZ IY	
5303	0510		TWLDK	
5304	5410		STOR6	
5305	5132		LDJ ACNT2	
5306	1000		MPY	/I X HIGH ORDER - DOUBLE PRECISION
5307	1302		LJKFRS	
5310	1450		CLR 0	
5311	4550		ADJ IY+1	/ADD IT (LOW)
5312	5551		STJ IY+1	
5313	1374		EXJK	

5314	1445	SIZ 0	/CHECK FOR CARRY
5315	1504	INC J	
5316	4556	ADJ IY	/ADD IT (HIGH)
5317	5557	STJ IY	
5320	3545	ISZ ACNT2	
5321	3147	DSZ ACNT1	/LAST?
5322	7145	XCT CJMP	/NO
5323	0640	TWJPS	/YES, FLOAT SUM OF Y
5324	5100	TPCHK	/CHECK FOR
5325	5062	LDJ POSST	/ TRIPLE PRECISION
5326	2304	SUBL 04	
5327	5460	STJ POSST	/FLOAT SUM OF I X Y
5330	0640	TWJPS	/MAY BE TRIPLE PRECISION
5331	5100	TPCHK	
5332	5055	LDJ POSST	
5333	2204	ADDL 04	
5334	5453	STJ POSST	
5335	5161	LDJ IDCON1	/START PART OF ROUTINE TO FIND SLOPE
5336	0540	TWSTJ	
5337	0402	0402	
5340	0640	TWJPS	
5341	0400	0400	/ENTER FPP
5342	0500	FTWLD	
5343	5231	ICON1	
5344	7406	FLOAT	
5345	0540	FTWST	
5346	5231	ICON1	
5347	0700	FTWMT	
5350	5244	Y	
5351	5437	FSTOR STOR6	/STOR6=(SUM I)*(SUM Y)
5352	0500	FTWLD	
5353	5247	N	
5354	7406	FLOAT	
5355	0540	FTWST	
5356	5247	N	
5357	0700	FTWMT	
5360	5240	IY	
5361	4027	FSUB STOR6	
5362	5431	FSTOR STOR2	/STOR2=N*(SUM IY) - STOR6
5363	0500	FTWLD	
5364	5231	ICON1	
5365	0700	FTWMT	
5366	5231	ICON1	
5367	5421	FSTOR STOR6	/STOR6=(SUM I)**2
5370	0500	FTWLD	
5371	5234	ICON2	
5372	7406	FLOAT	
5373	0700	FTWMT	
5374	5247	N	/ACCUMULATOR=(SUM I*I)*N
5375	4013	FSUB STOR6	
5376	0700	FTWMT	
5377	5037	DEPP	
5400	5410	FSTOR STOR6	/STOR6=DEPP*((SUM I*I)*N-(SUM I)**2

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5401	5012	FLOAD	STOR2	
5402	6406	FDIV	STOR6	
5403	5413	FSTOR	SLOPE	/SLOPE=STOR2/STOR6
5404	1001	FEXIT		
5405	0620			
5406	5146	[RETURN	LSR	
5407	5244	POSST,	Y	
5410	0000	STOR6,	0000	/TEMP STORAGE
5411	0000		0000	
5412	0000		0000	
5413	0000	STOR2,	0000	
5414	0000		0000	
5415	0000		0000	
5416	0000	SLOPE,	0000	
5417	0000		0000	
5420	0000		0000	

SE 6203

A1	=	4072
ACNT1	=	5252
ACNT2	=	5253
AD1	=	5071
ADDR	=	4052
ADR1	=	4053
ADR5	=	4672
ADR6	=	4767
B	=	5042
BUFF	=	4566
BUFRT	=	4575
CJMP	=	5255
CNT1	=	4033
CNT2	=	4034
CNTR	=	4574
CON	=	4015
CON1	=	4440
CON2	=	4443
CON3	=	4446
CON4	=	4451
CON5	=	4467
CONST1	=	4714
CONTN1	=	5257
CONTN2	=	5163
CT	=	4167
CTM	=	4257
DAT1	=	4152
DEPP	=	5037
DFRM	=	4322
DT0	=	4335
E	=	5034
E1	=	4156
E2	=	4164
EFP	=	5045

APPENDIX A

EN1	= 4772
EN2	= 4774
ENE	= 4221
EXJK	= 1374
FCLR	= 1004
FDGE	= 4206
FJMP	= 6000
FLOAT	= 7406
FLUX	= 4464
FNEG	= 1003
FPCON1	= 4730
FSIN	= 1006
FSIP	= 1007
FSQRT	= 7401
FUDG	= 4354
GF	= 4200
GT	= 4203
H2CO	= 4274
HC	= 4172
ICON1	= 5231
ICON2	= 5234
IDCON1	= 5254
IFIX	= 7405
INCR	= 4561
INIT	= 4770
INPT	= 4071
INST	= 4572
IOST	= 4371
IY	= 5240
LSR	= 5146
MASK	= 4646
MESSG	= 4051
MEW	= 4306
MW	= 4175
N	= 5247
NPTS	= 5230
OMR	= 4035
OMR1	= 4041
OPT	= 4473
OUST	= 4462
P	= 5031
P1	= 4153
P2	= 4161
PFSET	= 4673
PFST	= 4570
POINT	= 4211
POS	= 5142
POS2	= 5141
POSST	= 5407
PREC	= 5136
PT1	= 4545
PT2	= 4617
PT3	= 4560

PT4	= 4654
PT5	= 4661
PTNO	= 4567
PTWTH	= 4571
RETN	= 4367
SIG	= 4470
SLOPE	= 5416
STOR1	= 4764
STOR2	= 5413
STOR5	= 5143
STOR6	= 5410
STR1	= 4454
STR2	= 4457
STT	= 4054
TITLE	= 4232
TPCHK	= 5100
TWO	= 4731
WI	= 4717
WIN	= 4734
WIP	= 4722
WMSQ	= 4725
XC2	= 5076
XCLR	= 5060
XMAX	= 1131
Y	= 5244
ER 0002	

APPENDIX B.—SAMPLE PROGRAM RUN

This appendix provides a sample run of the computer program BASC-12.

DERIVATIVE PROGRAM
POINT =1 ENERGY =1.1
POINT =2.0 ENERGY =.2E+01
COUNT TIME =1.E+03 H CONCEN=.1E-02
MECH WIDTH=.1
IN GOUP =1 OUTPUT GROUP =2
WI MULT= 3.

* DERIVATIVE PROGRAM
POINT =1 ENERGY =1
POINT =2 ENERGY =2
COUNT TIME =1 H CONCEN=1
MECH WIDTH=.1
IN GOUP =1 OUTPUT GROUP =2
WI MULT= 3.

* 0

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